SoftwareX 25 (2024) 101635

Contents lists available at ScienceDirect

SoftwareX

journal homepage: www.elsevier.com/locate/softx

Original Software Publication

TeamWellX: Leveraging genetic algorithms to optimize team health and well-being toward sustainable game development and strategy

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ARTICLE INFO

Keywords:
Roster-formulation
Genetic algorithm
Fitness-function
Machine learning
Team strength

ABSTRACT

Building a well-rounded roster is crucial in sports, as it directly impacts a team's long-term success. However, the process can be complex, with various factors to consider, such as team dynamics, player skills, and financial limitations. This study aims to develop effective strategies for constructing teams in different sports, starting with American Football in the NFL and potentially expanding to other similar disciplines. In this paper, we propose an innovative tool that leverages genetic algorithm, signifying a method that balances between overall team positional strength, fosters collective team success, and encourages team's long-standing well-being. This data-driven approach also presents a generalizable solution that can be employed across similar team sports, transforming the pattern of roster optimisation strategies and team's performance augmentation.

Metadata

Nr	Code metadata description	*Please fill in this column*
C1	Current code version	v1
C2	Permanent link to code/repository used for this code version	https://github.com/efox2/NFL-Team-Construction
C3	Permanent link to reproducible capsule	https://github.com/efox2/NFL-Team-Construction
C4	Legal code license	MIT License
C5	Code versioning system used	https://github.com/
C6	Software code languages, tools and services used	R
C7	Compilation requirements, operating environments, and dependencies	RStudio
C8	If available, link to developer documentation/manual	https://github.com/efox2/NFL-Team-Construction/blob/main/README.md
C9	Support email for questions	sumedhk@buffalo.edu

1. Motivation and significance

Most of the major team sports featured in the United States cover the province of analytic probability that can profoundly influence strategies, personnel decisions, and game outcome, guaranteeing overall success of a team. Amongst the diverse substances a pivotal component resides in roster construction– the development of a listing of players available for a specific team. The description can also be supported by the portrayal in the movie "Miracle", where the U.S. men's Olympic hockey team head coach Herb Brooks states, "I'm not looking for the best players, Craig. I'm looking for the right ones."[1]. This testimony underlines roster construction's ubiquitousness across abundant team sports, justifying that even a team constituting an apparently subordinate players lineup can outshine an icon-loaded team that seems invincible on paper. Also, some sports have multiple franchise leagues, overloading players throughout the year. Footballers must play both for their franchise league teams and national teams. Cricketers must play in three different formats: test, ODIs, and T2O, for their national team. In this complicated situation, the management team of any sport struggles to maintain or find the sweet spot for players' workload and effective team combination toward a winning cause. Explaining further, we proceed answering the critical inquiry: How can we construct the optimal roster for the impending season? What identifies the upright collection of players and what are the factors that influence a team's triumph?

In a team selection process, earlier manual procedures had many

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https://doi.org/10.1016/j.softx.2024.101635
Received 1 October 2023; Received in revised form 19 December 2023; Accepted 14 January 2024
Available online 30 January 2024
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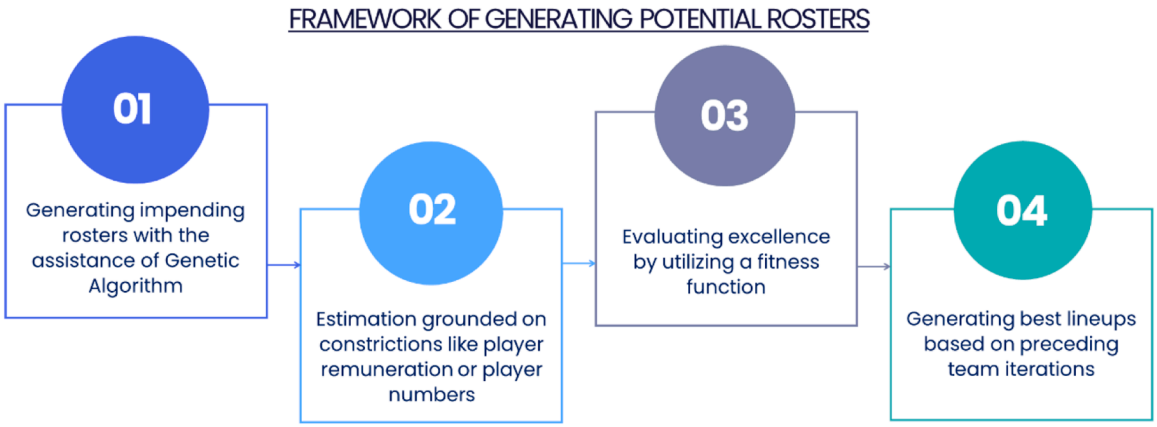

Fig. 1. Flow chart outlining primary functions of the TeamWellX software.

drawbacks such as biasness related to personal liking and disliking, personal grudges between selection committee and players as well as social and political pressures leading to failure and disappointment. To avoid this practice many software-based effective solution has been proposed lately that can give an optimistic result in a lesser amount of time. However, many of these conventional solutions have made it difficult to achieve statistical success in the sphere of sports analytics [2]. One of the major limitations include they do not cover all the aspects required such as accounting only on the overall team quality as the sole base for a team's success. For instance, one study has utilized a genetic algorithm with a standard fitness function without considering the individual performance of players for cricket team formation [3]. This problem-solving system lacks depth by stating that the technology provided should have considered all significant factors to make it more accurate. To choose the optimal selection of players from a roster, a study illustrates a method of integrating genetic algorithms and decision trees [4]. Another research examines NFL team success determinants using the Poisson regression model to perceive the competent way for acquiring skillful players and revealing ways to improve NFL player performance [5,6]. In our research, we propose augmenting these approaches through a learned fitness function rooted in machine learning (ML) techniques. It methodically allocates scores symbolic of player accomplishment potential such as rankings, spots, and prior performance measurements to iteratively assess roster development. We initiate by employing player scores from the popular video game Madden for the American Football, allowing us a singular metric for measuring player quality. By reflecting the selected variables, a weighted average is estimated for each team where the fitness function selects the roster with the highest weighted mean [7]. Moreover, to accommodate possible changeability in player salaries, we presume that player compensation remains constant– a useful assumption for our initiative.

According to our analysis, combining two prevalent approaches– prediction of best performing players and optimization of roster composition based on their positions– could improve team construction analytics [8]. Thus, our endeavour will prioritize high-ROI players alongside their comparable scores and ranking. Employing a ML-based fitness function, our conceived genetic algorithm decides the optimal roster formation within a set of predetermined constraints. This is accomplished by initializing the roster with a programmed number of players and permitting the genetic algorithm to pick the remaining players using alterations of numerous iterations until the target is attained. Our plan could be applied to other multi-player sports that require a more inclusive team selection strategy partaking the blend of machine learning and evolutionary algorithms into a more organized and applicable team-building tool. This projected innovative system will poise the expansion of an array of team-building strategies, as well as will revolutionize the field of sports analytics —be it in the preseason or for individual transactions during the current season– thereby providing teams with a more effective tool maximizing financial resources for teams in specific contextual demands.

The manuscript focuses on the following distinct sections. The proposed software architecture, involving its features and operations are clarified in Section 2. In Section 3, illustrative examples are offered along with discussing how this application grant improvements for the associated stakeholders. The extensive scientific impact is outlined in Section 4. Ultimately, Section 5 concludes by summarizing our endeavors and outcomes.

2. Software description

As mentioned earlier, the framework employs a Genetic Algorithm to independently reveal the optimal player configuration for a football team, accounting for a multitude of parameters and correlations between them. Therefore, this tool will aid team management personnel in meticulously devising and forging the most advantageous roster for their National Football League Team, minimizing operational expenses, and expanding team competitiveness.

Genetic algorithms (GA) are omnipresent across a variety of applications, for instance NCAA college football position techniques [9], cricket team configuration [10], and scheduling [11]. In biological scenario, individuals from a population compete for survival, compelling the accretion of constructive attributes across generations. In machine learning, genetic algorithms are an adaptive algorithm which imitate this notion of natural genetics when employed to data, upon encountering two specific prerequisites. Firstly, the target challenge of interest must be appropriately denoted either through binary encoding or bit string. Secondly, encoded results compel a well-organized fitness function to quantitatively determine their effectiveness. These algorithms exhibit superior intellect compared to arbitrary ones as they employ historic data to examine the search towards the finest performing state in the solution space. Since GA can uncover the solution in each challenge with multiple subgroups simultaneously, it is contemplated to be appropriate for roster formulation [12,13].

Fig. 1 shows a representation of our proposed tool's operating structure. Initially, prospective rosters are generated using the genetic algorithm. Consequently, these rosters are exposed to a comparison against our preconceived restraints, penalizing team constructions that violate number of players or salary restrictions. Afterwards, the assessment of the team's quality is identified through a fitness function (simple or machine-learned). Finally, the top teams are selected, forming the foundation to produce new rosters based on the top teams from the previous iteration. This iterative procedure is repeated until there is no further roster improvement. The impact of player scores and positions

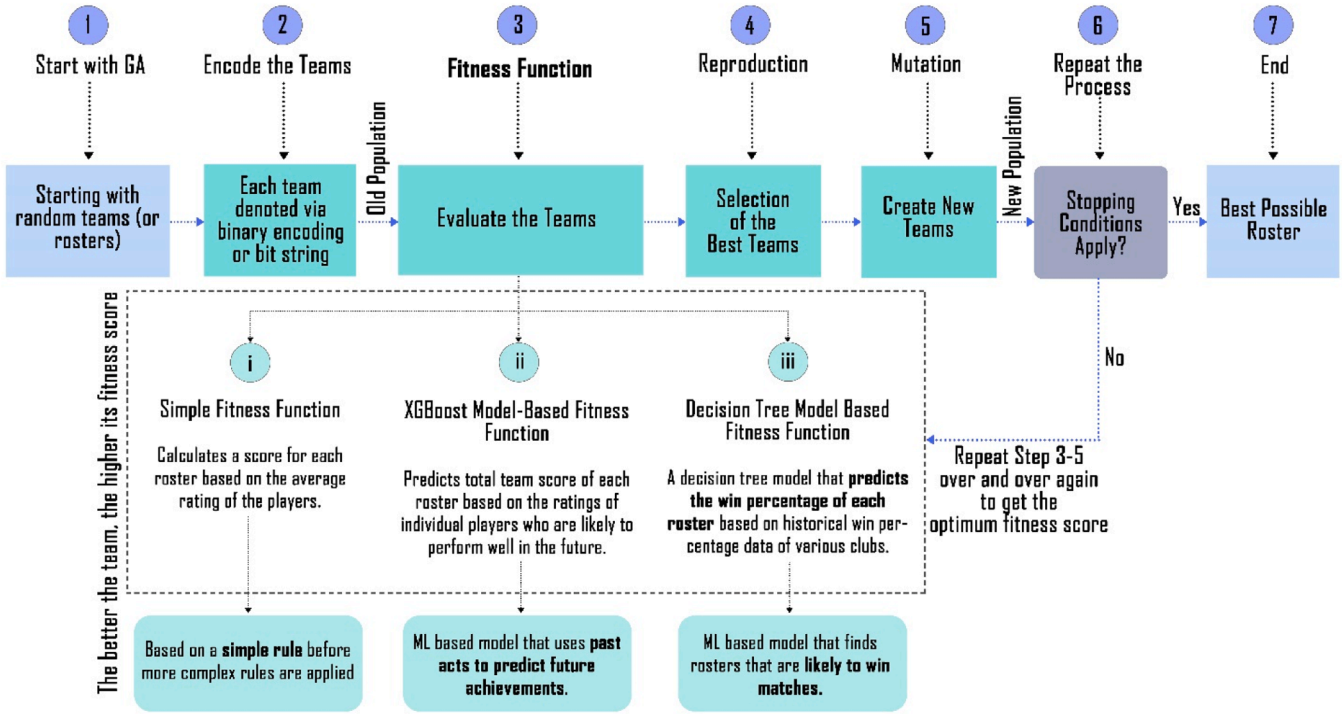

Fig. 2. Simplified visual representation of Genetic Algorithm's flow process.

on team performance was found to be statistically significant in each season, as determined by a correlation analysis of the dataset variables. The measure of team performance was based on the variable 'Team Score'. To comprehensively account for the intricacies inherent in the dynamic nature of player statistics, a greater number of seasons' worth of data was necessary. However, the variable denoting 'Team Score' exhibited instances of missing data. Thus, a semi-supervised ML model was developed to forecast the 'Team Score' through the extreme gradient boosting algorithm, XGBoost and decision tree regression.

XGBoost is an ensemble-based boosting technique that surpasses in handling missing data and offers extreme model performance capabilities by assessing every single bit of the data values in each database, resulting in models that can outperform other ML models with higher accuracy, higher precision, lower false positives, and lower missing rates, even for large datasets [14,15]. However, overfitting in XGBoost can appear, leading to high training accuracy but lower test accuracy and inaccurate predictions of new data. To address this, adding randomness to make training resilient against noise could help [16]. Additionally, imbalanced data distribution can impact the training of XGBoost model [17]. Hyperparameter tuning can appear as a solution to receive better model performance by utilizing methods like grid search, Bayesian optimization, and random search [18]. Our ongoing research seeks to address these issues to further progress the overall performance and forecast the accurate predictions.

As mentioned before, the genetic algorithm exhibited three definite iterations, each utilizing a fitness function to assess the roster:

- *Simple Fitness Function:* The initial genetic algorithm adopts a simple fitness function, which involves the computation of a weighted mean rating resulting in a balanced allocation of resources between defensive and offensive strategies, as well as a prioritization of key positions such as the quarterback. The roster is selected by this function based on its metric value. This approach may aid in

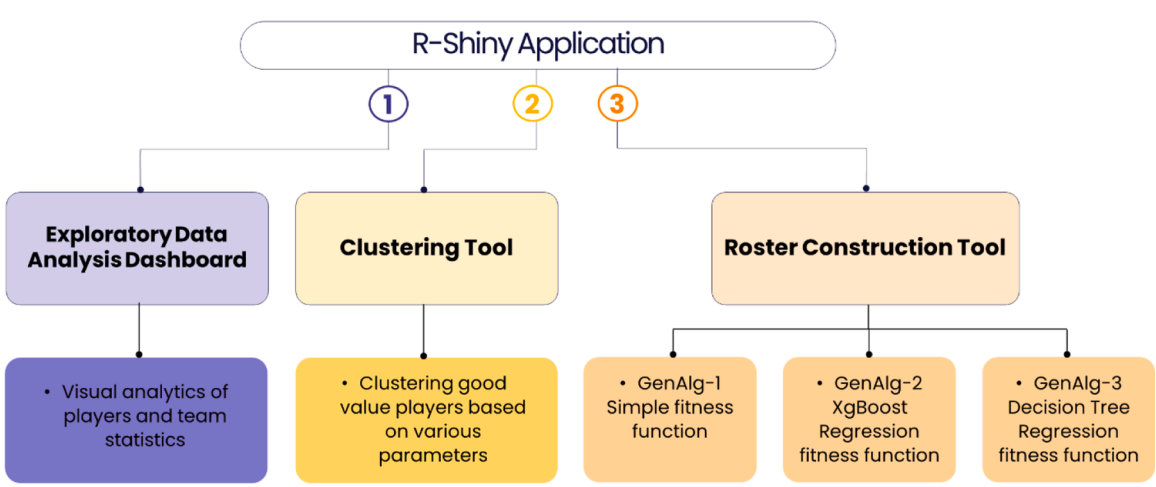

Fig. 3. TeamWellX salient features considering NFL roster.

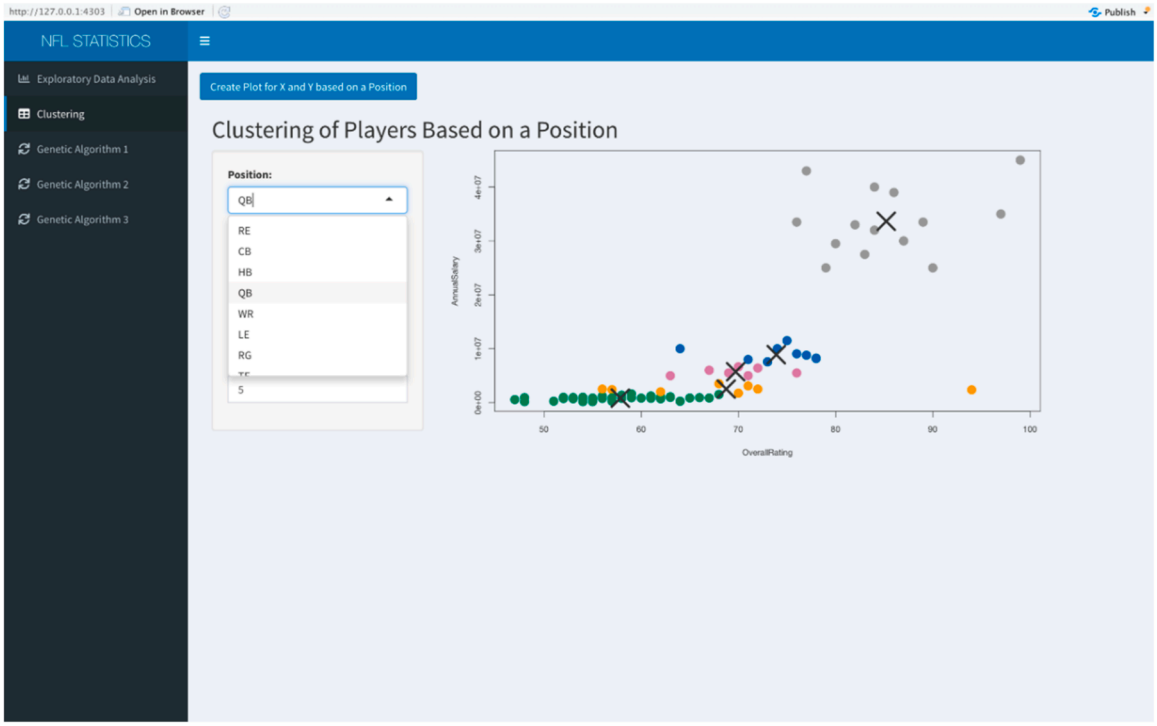

Fig. 4. Clusters to group players based on user requirements.

selecting the optimal roster prior to the development of a machine learning-based fitness function [19].

- *XGBoost model-based fitness function:* The XGBoost machine learning model has been employed to forecast the aggregate team score of the produced roster by leveraging the ratings of each individual player within the roster [20].
- *Decision Tree Model based fitness function:* The Decision tree model forecasts the win percentage of the generated roster, applying the historical win percentage data of various clubs. The predicted value subsequently assesses the roster generated by the genetic algorithm [21].

The operators involving the collection, recombination, evaluation, mutation, and iteration functions collectively concerning solution convergence, as illustrated in Fig. 2.

Our outcomes indicate that the ML-based fitness function demonstrated superior performance in comparison to its basic counterpart. Though applicable to numerous team sports, in this paper, we show illustrative examples considering NFL (American professional National Football League). Discovering the perfect combination for a 53-man NFL roster is a complex task. The many possibilities make it impossible to analyse thoroughly using a straightforward approach. We assessed subjects by their convergence rate, the number of generations needed to find the best solutions, and the overall value of the selected genes.

3. Illustrative examples

As previously mentioned, the R Shiny application presents three critical functions for scrutinising player information and for classifying high-value players established according to their ratings and remunerations [22] (Fig. 3).

3.1. Exploratory data analysis dashboard

This feature empowers management team to make informed decisions by granting them a platform to explore related records. For instance, it aids users to investigate diverse statistics pertaining to NFL teams and their pXG layers such as:

Table 1
Previous team composition.

Position	Number of players
Quarter Back	2
Running Back	4
Offensive Linesman	9
Tight End	3
Wide Receiver	6
Defensive Lineman	9
Line Backer	7
Defensive Back	10
Kicker	1
Punter	1
Long Snapper	1

Annual Salary vs. Overall Rating: This section depicts the correlation between annual salary and the overall rating corresponding to player's ranking. Remarkably, an evident inclination arises— higher the overall rating of a player, higher is the annual salary. User can filter their search by filtering players based on their position, and team zone.

Value for Money: A bar plot is generated, representing the top 20 clubs that have maximized their remuneration per overall rating point. The highlighted bars represent the clubs with an overall rating exceeding 70 %- an indicator of having a beneficial roster. This imaging serves to analyse how the roster is composed across different clubs according to player ratings.

3.1.1. Clustering assessment for best value players

This tool clusters players based on their overall rating and annual salary, permitting users to pin down players presenting the best value. Afterwards, the user can deliberately select players for their team grounded on these criteria to establish a remarkable yet cost-effective team [22]. Fig. 4 shows how management team can filter specific positions to refine their search.

Table 2
Establishment of a best possible team configuration.

Position	Number of players
Quarter Back	3
Running Back	4
Offensive Linesman	9
Tight End	3
Wide Receiver	6
Defensive Lineman	9
Line Backer	7
Defensive Back	10
Kicker	1
Punter	1

3.1.2. *Genetic algorithm for ideal roster generation*

TeamWellX, the RShiny app, generates optimal club rosters employing genetic algorithms, following constraints like existing players and wage limit. The optimal roster is determined by the algorithm based on the outcomes from the three previously discussed different fitness functions. The first function computes the sum of all players' scores. Existing algorithms fall short in mitigating the trade-offs between satisfying several constraints. Table 1 illustrates the structure of a 53-man roster in the NFL [23]. Our modified algorithm concentrates on providing the optimum outcome by attaining an equilibrium amongst various restraints. Table 2 delineates the ideal team composition based on playing positions and ideal team composition, not the actual players. The actual player candidates associated with these positions could vary. We introduced an alteration to the outline– the Long Snapper (LS) position was eliminated because it is highly specialized and may not be relevant to most NFL teams. This modification presumes that an existing player could handle the LS role along with their designated position. By eliminating the LS position, we made space for an additional Quarter-Back position while maintaining 53-total players in both cases [23]. Fig. 5 demonstrates the algorithm's output, where the green emphasized are the recommended players for the roster construction.

The second genetic algorithm employs an XGBoost regression machine learning model to predict the overall score of a roster. The fitness function then leverages this score as the basis to evaluate all generated team scores. Fig. 6 indicates the players selected by the genetic algorithm using an XGBoost model as the fitness function. The third genetic

	Player Name	Team	Player Team Zone	Position	Overall Rating	Age	Annual Salary
17	Fletcher Cox	Philadelphia Eagles	NFC East	DL	96	29	17100000
544	Travis Benjamin	San Francisco 49ers	NFC West	WR	74	30	1075000
892	Sam Sloman	Los Angeles Rams	NFC West	K	68	22	825000
1032	Matt LaCosse	New England Patriots	AFC East	TE	66	27	293000
401	Adam Butler	New England Patriots	AFC East	DL	76	26	3750000
322	Cam Newton	New England Patriots	AFC East	QB	78	31	8250000
1602	Brian Lewerke	New England Patriots	AFC East	QB	54	23	713500
1574	Korey Cunningham	New England Patriots	AFC East	OL	56	25	920000
468	Chase Winovich	New England Patriots	AFC East	DL	75	25	956828
560	Kyle Dugger	New England Patriots	AFC East	DB	73	24	2082976

Fig. 5. Roster selected with a kicker as one of the 3 predicted players.

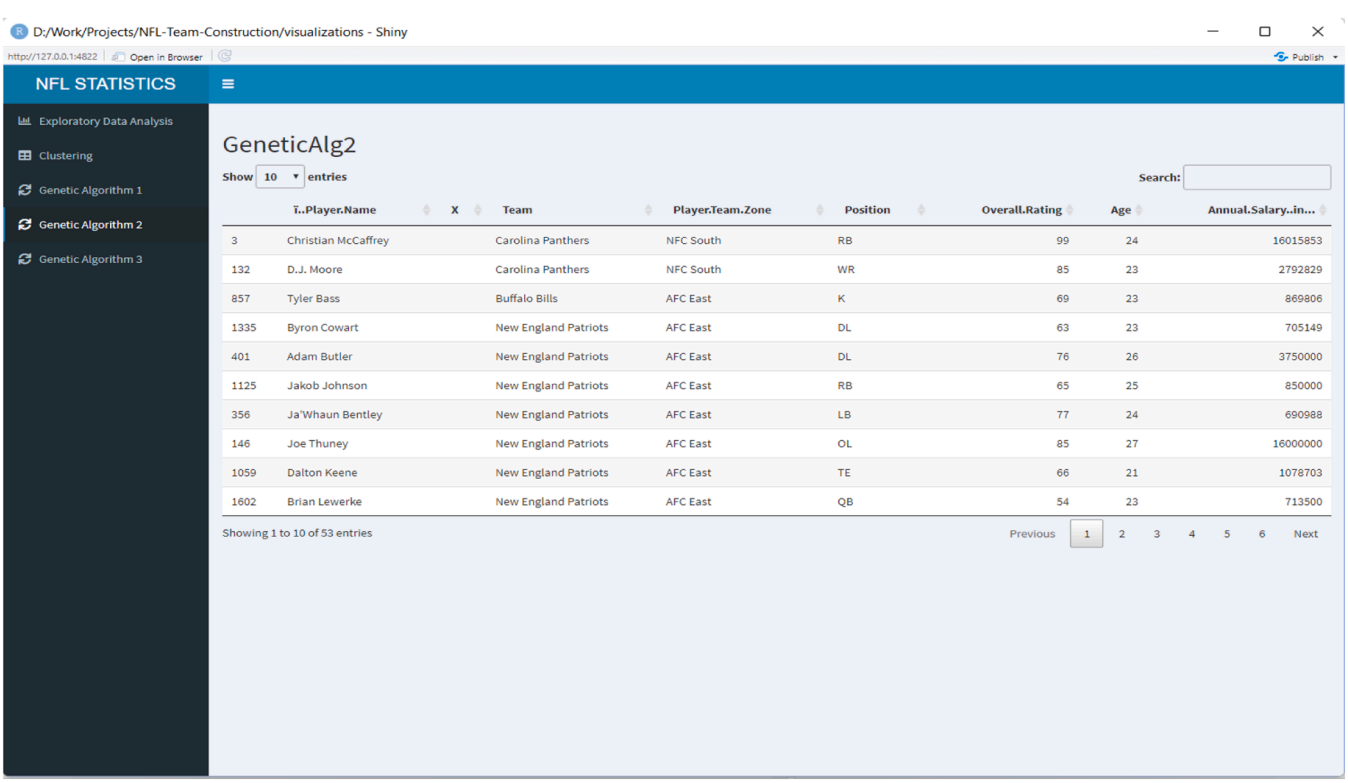

Fig. 6. Genetic algorithm type 2 result based on the XGBoost regression model.

Table 3
Comparison between the models.

S. No	Model	Number of Iterations
1	Genetic Algorithm	90
2	Importance to player positions (XGBoost)	85
3	Importance to individual player win percentage (Decision Tree)	70

algorithm employs a Decision Tree Regression model to estimate the overall winning percentage of a given roster. The percentage is afterwards utilized to determine whether the generated teams are suitable or not. Table 3 provides a comparison between these models and number of iterations required to reach the first acceptable solution.

4. Impact

Conventional algorithms such as Greedy Search and Brute Force encounter challenges to offer an appropriate solution for optimization problems and are characterized as time-consuming due to their concentration on single peak objective functions. Conversely, genetic algorithms consider an array of feasible solutions, going beyond finite possibilities and spanning across compound dimensions. Our innovative and collective approach, along with its prospective benefits allows researchers, players, and team managers to align their own interests, leveraging the rational and straightforward nature of this framework. From an administrative perspective, this algorithm expands its functionality to team managers, talent acquisition managers, and headhunters in recruiting and retaining employees. Furthermore, the software's visualizations of player clustering can help identify high-performing players with moderate salaries– a rigorous feature to maximizing player value and improving team performance within the domain of sports and health analytics [24]. It may empower franchise executives in reducing expenses. Exploratory data analysis continues its significance to investors, marketers, and associated stakeholders by unearthing insights and future opportunities for them. From a sports analytics perspective, recruiters can also utilize algorithmic performance comparisons through their active participation to advance robust frameworks for player selection. During post-simulation of roster formation, the program may facilitate in player, team, or zone collection. The app's capacity to identify high-value players within budget has the potential to revolutionize team management and decision-making in the sports industry. The course of our development plan also offers scope for advancement including modification of the XGBoost algorithm to consider additional relationship joining player positions and player statistics constraints.

5. Conclusion

This paper represents a framework propositioning an optimal roster formation generated through a combination of machine-learned fitness function and genetic algorithm preferring significance direction to stakeholders, assisting them in planning their hiring and retention strategy for the upcoming season. In addition, it focuses on strategies that franchises can perform to prepare their players for success by controlling financial factors and assembling the best available pool of players on an ideal roster. This introductory concept holds capacity for future expansions, by individualizing rosters for diverse teams. Given that certain common criteria contribute weighted effect, rather than absolute impact on overall team success, a team-based software recommendation could be a stimulating avenue for further exploration. If team specifications and circumstances are known in advance, constraints for that team could be automatically loaded into our algorithm, enhancing its pertinency.

CRediT authorship contribution statement

Sumedh Khodke: Conceptualization, Data curation, Formal analysis, Writing – original draft. **Evan Fox:** Data curation, Software. **Nafisa Anjum:** Visualization, Writing – original draft, Writing – review & editing. **Dineshkumar Katta:** Methodology, Software. **Amritha Subburayan:** Resources, Writing – original draft. **Sanjay Aravind Loganathan Ravichandran:** Visualization. **MD Ahasan Habib:** Validation. **Nazmus Sakib:** Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

We acknowledge the Madden NFL website as the crucial data source for this project. Data collected from the site was pre-processed to extract player ratings based on position, salary, age, and statistics to create team ratings for 53 ideal players, as shown in Table 1. Our machine learning model uses this roster composition for positional comparison with Madden ratings as the ground truth. Data source: https://www.ea.com/games/madden-nfl/player-ratings.

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